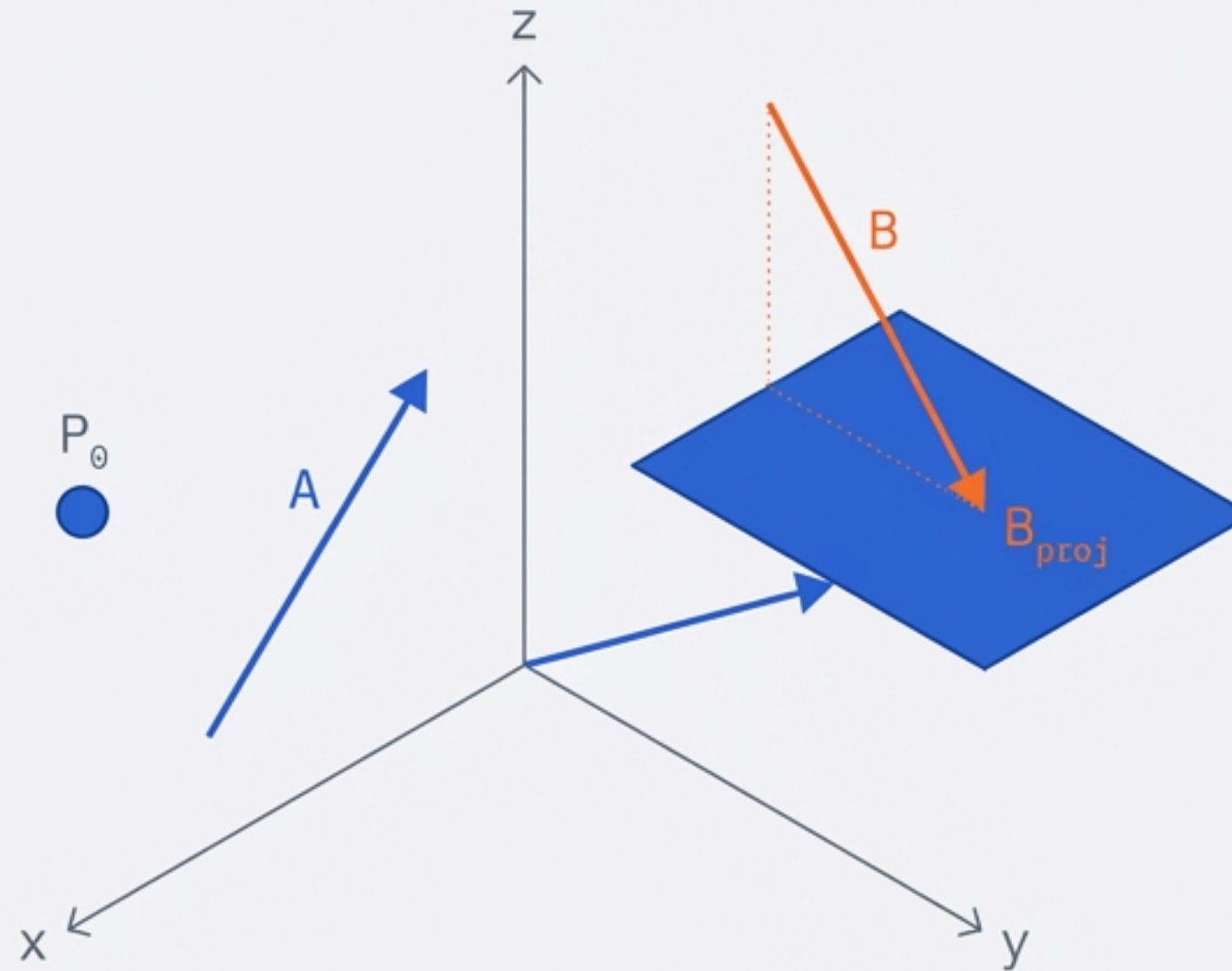


The Geometry of Data

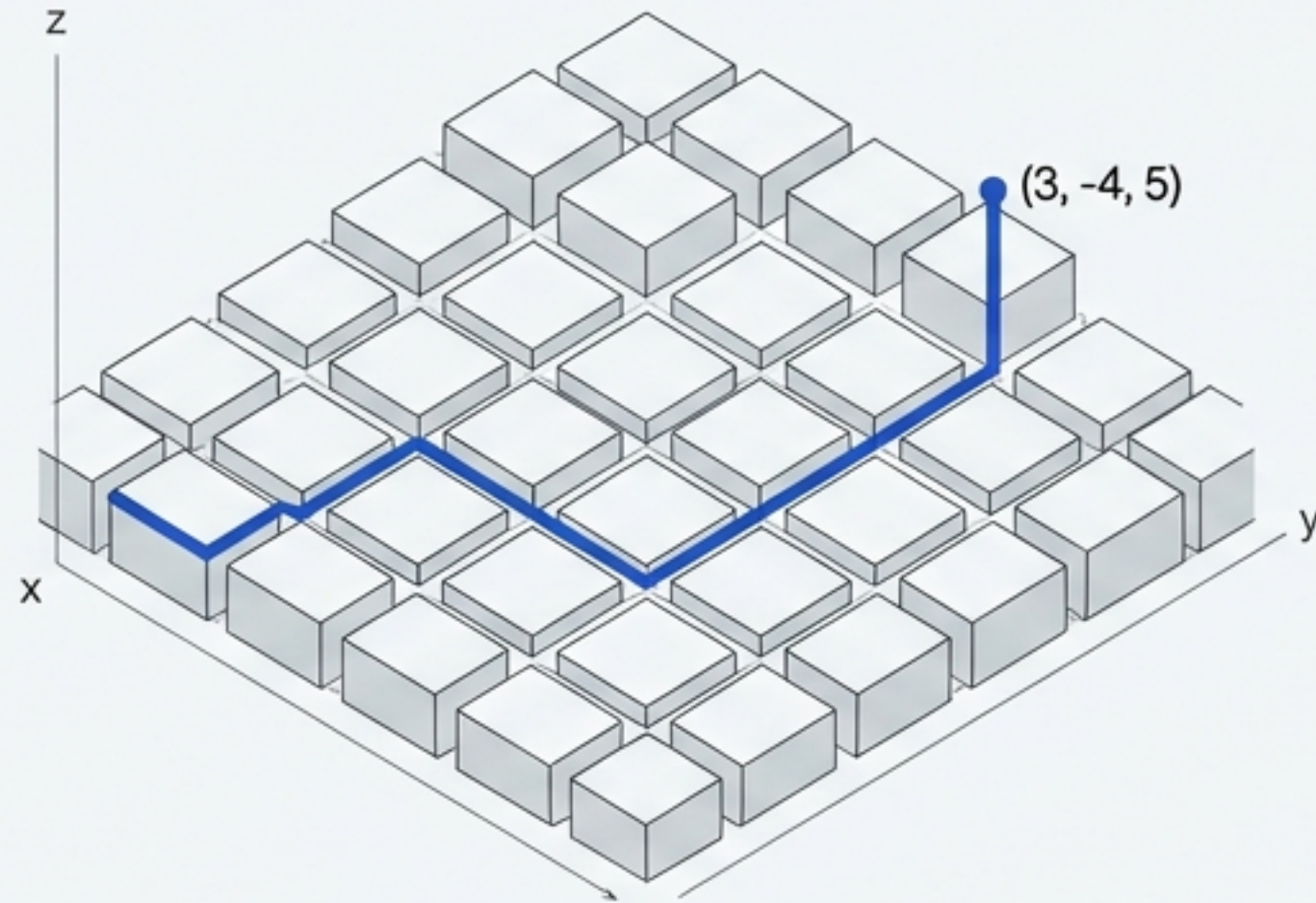
A Visual Guide to Vector Operations & Mechanics



From fundamental spatial components to complex dimensional interactions.

Defining Magnitude: The Norms

Norm-1 (Manhattan / L1)

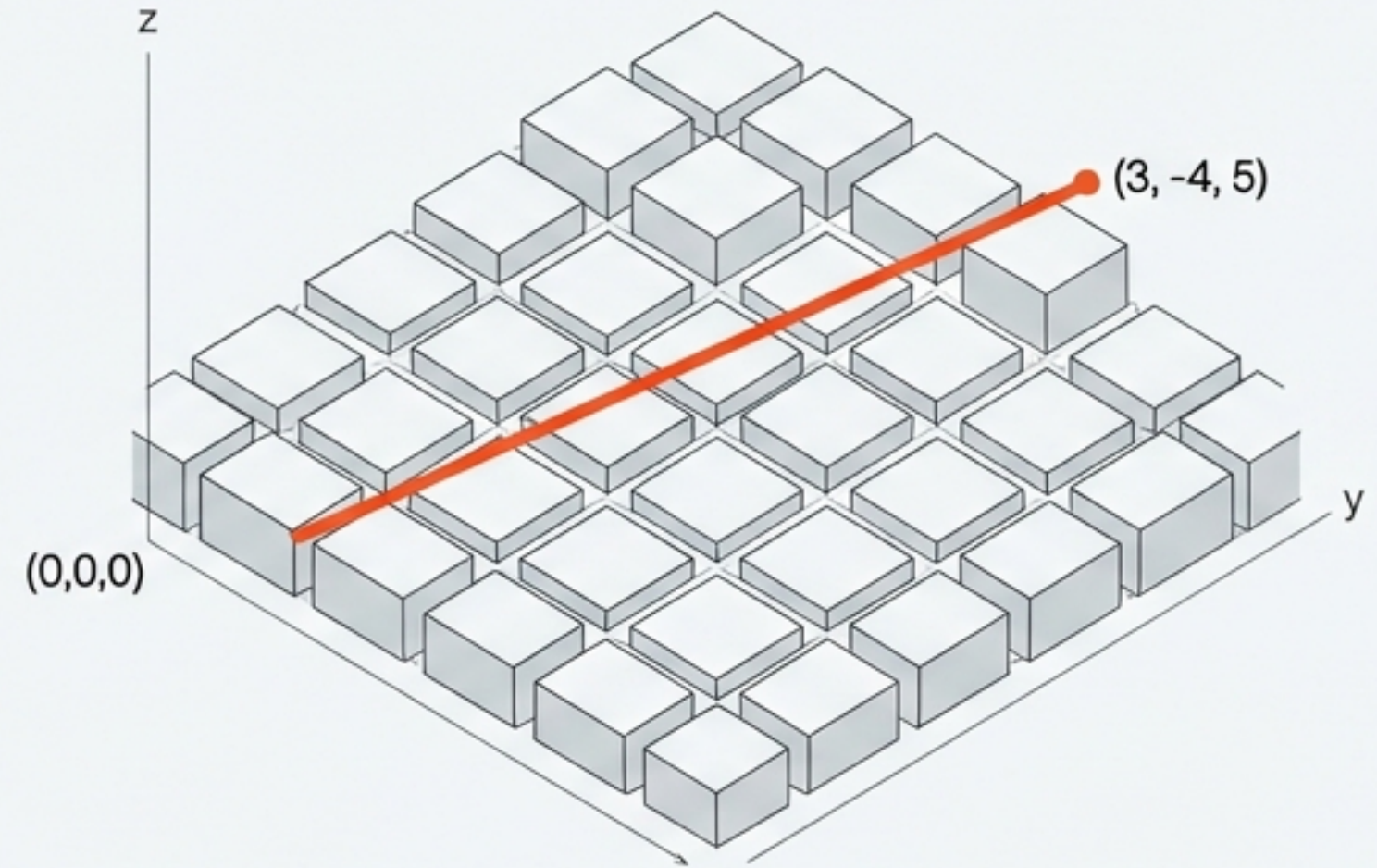


Formula: $\|x\|_1 = \sum |x_i|$

Context: Measures distance along grid-like paths. Sum of absolute values.

Example: $x = (3, -4, 5) \rightarrow 3 + 4 + 5 = 12$

Norm-2 (Euclidean / L2)

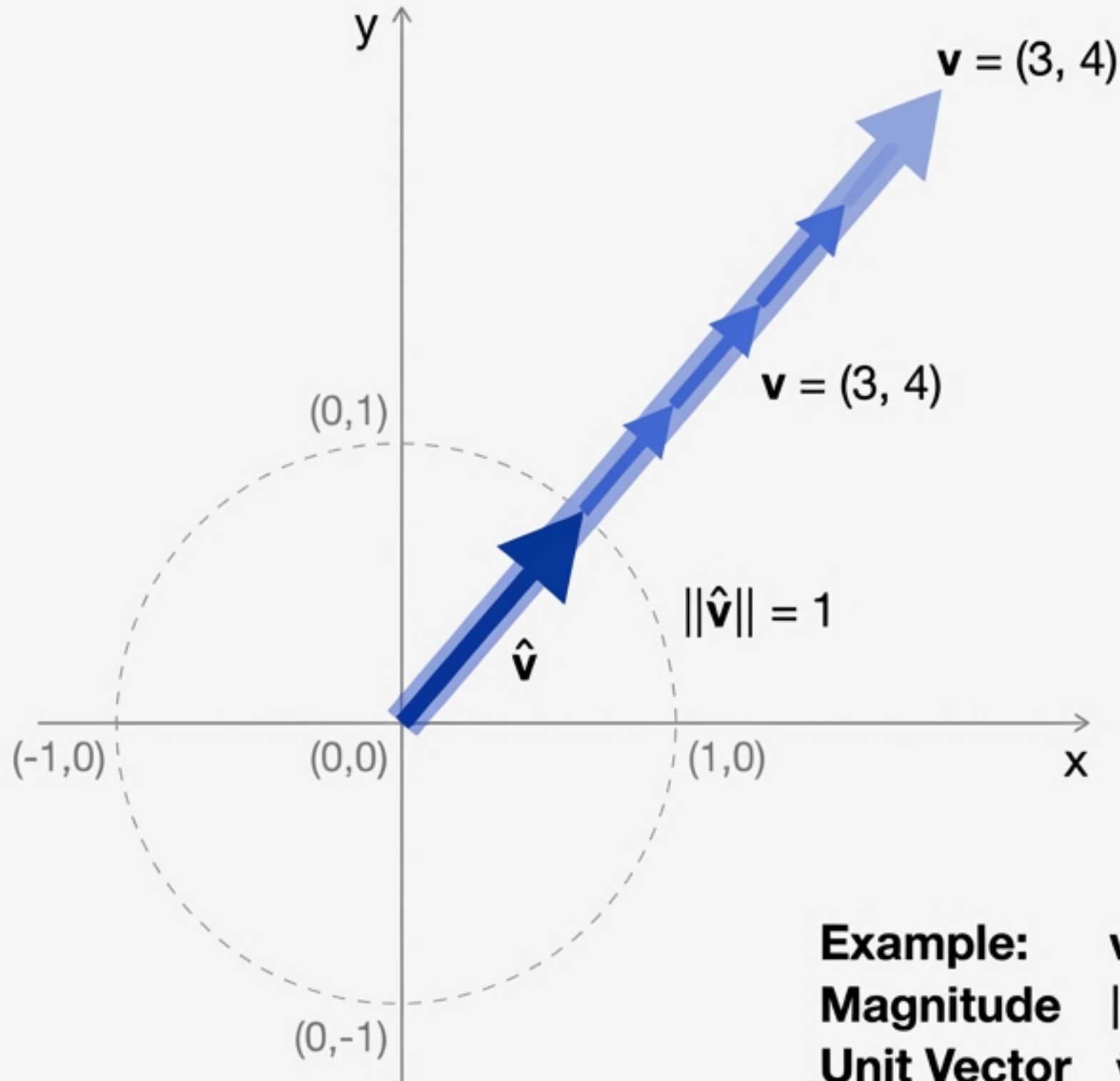


Formula: $\|x\|_2 = \sqrt{\sum x_i^2}$

Context: The 'usual' straight-line distance. Root of sum of squares.

Example: $x = (3, -4, 5) \rightarrow \sqrt{(9+16+25)} = \sqrt{50} \approx 7.07$

Pure Direction: The Unit Vector



Concept: A Unit Vector has a magnitude (length) of 1 but retains the original direction. Denoted by a 'hat' symbol ($\hat{\mathbf{v}}$).

The Formula:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

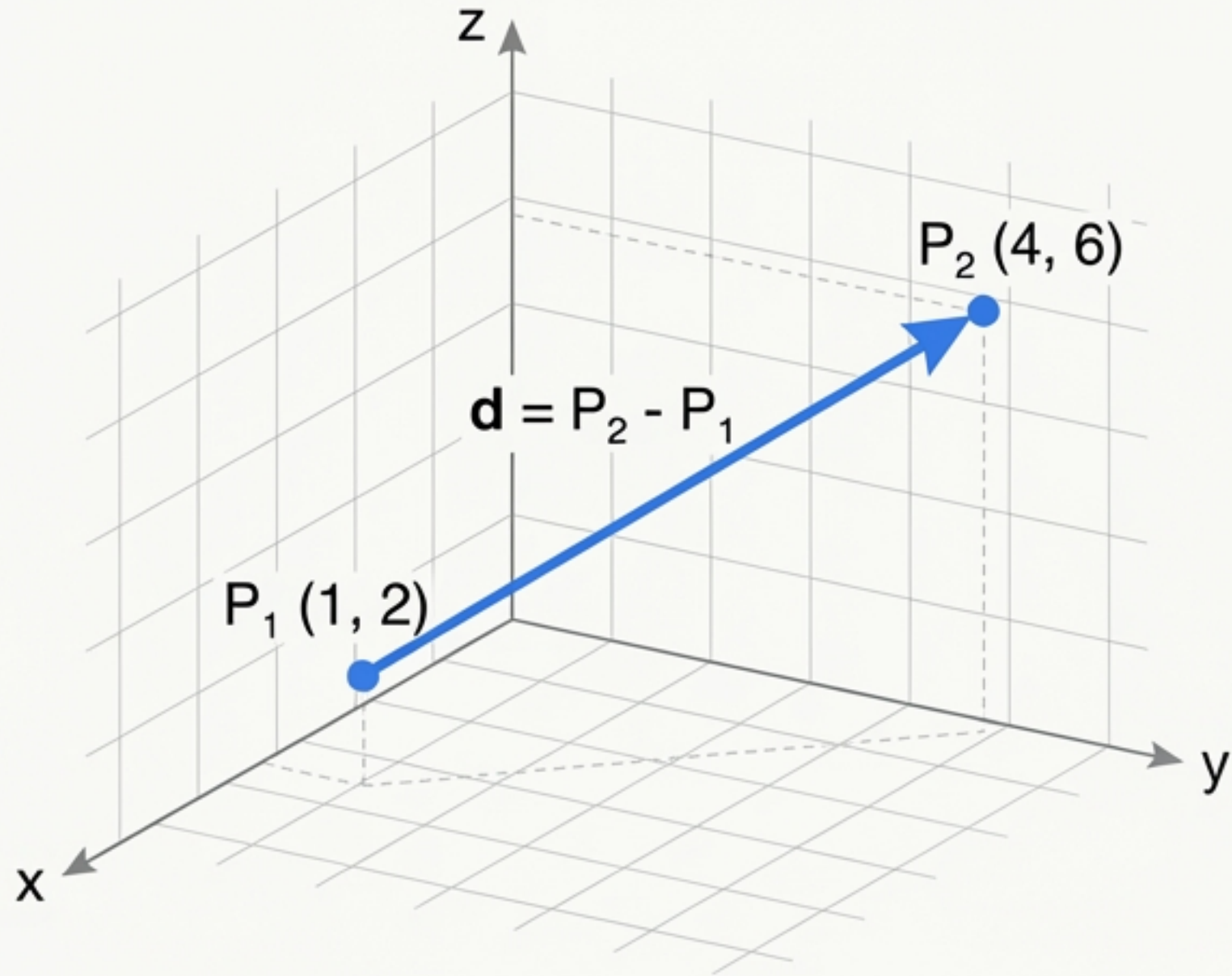
(Dividing the vector by its own magnitude)

Why it matters in Data Science:

- **Normalisation:** Scaling inputs to a standard range.
- **Cosine Similarity:** Vectors are normalised before angle computation.
- **Gradient Descent:** Represents the direction of steepest descent.

Example: $\mathbf{v} = (3, 4)$
Magnitude $\|\mathbf{v}\| = 5$
Unit Vector $\hat{\mathbf{v}} = (3/5, 4/5) = (0.6, 0.8)$

Distance in Space: The Difference Vector



Core Logic: The distance between two points is derived from the magnitude of the difference vector.

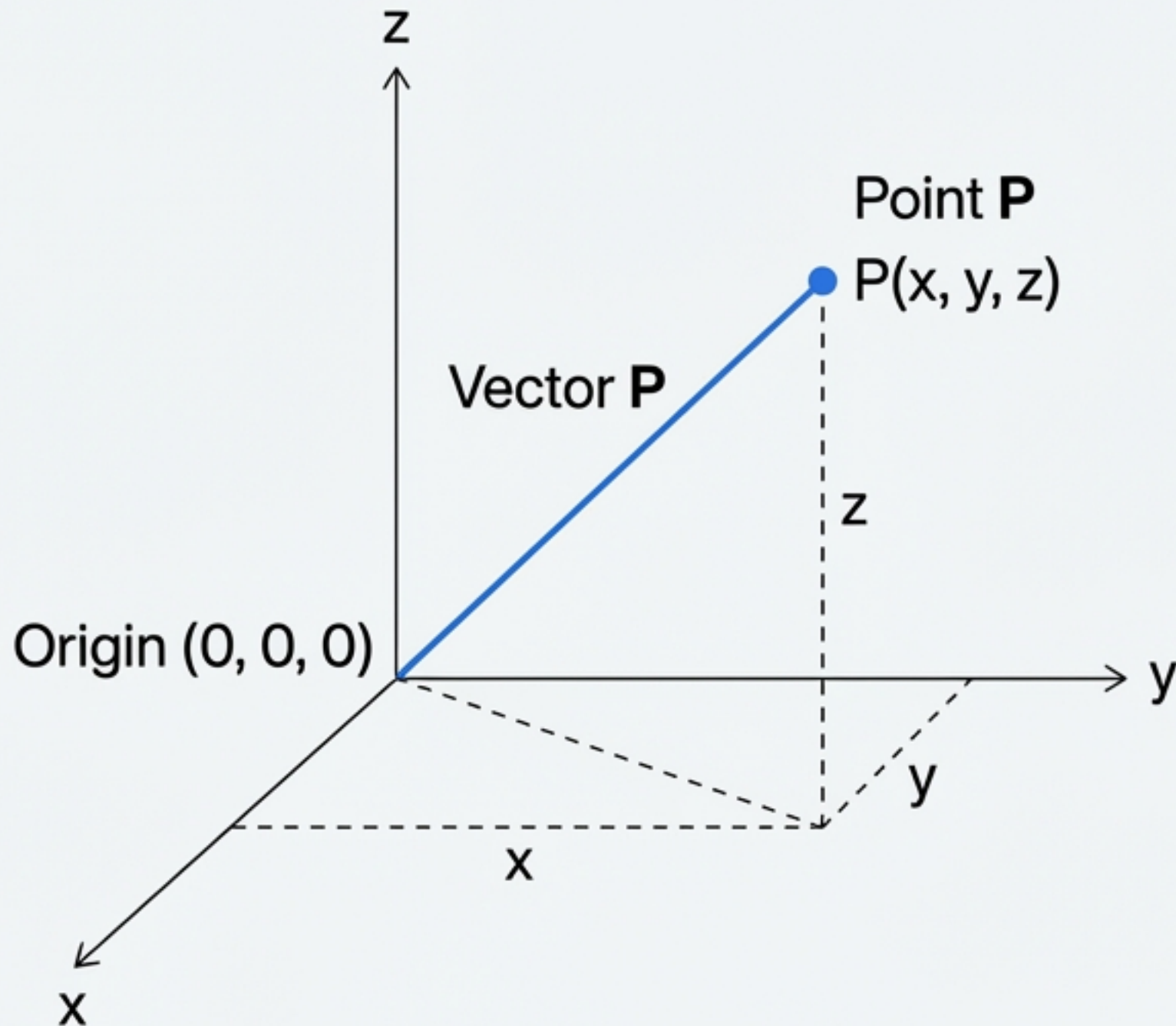
Calculation:

1. Points: $P_1(1, 2)$ and $P_2(4, 6)$
2. Difference: $d = (4-1, 6-2) = (3, 4)$
3. Magnitude: $|d| = \sqrt{(3^2 + 4^2)} = 5$

General Formula:

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Position as a Vector



If a point is viewed as a vector $\mathbf{P}$, the distance from the origin is simply its Magnitude.

Formula:

$$d = \sqrt{x^2 + y^2 + z^2} = |\mathbf{P}|$$

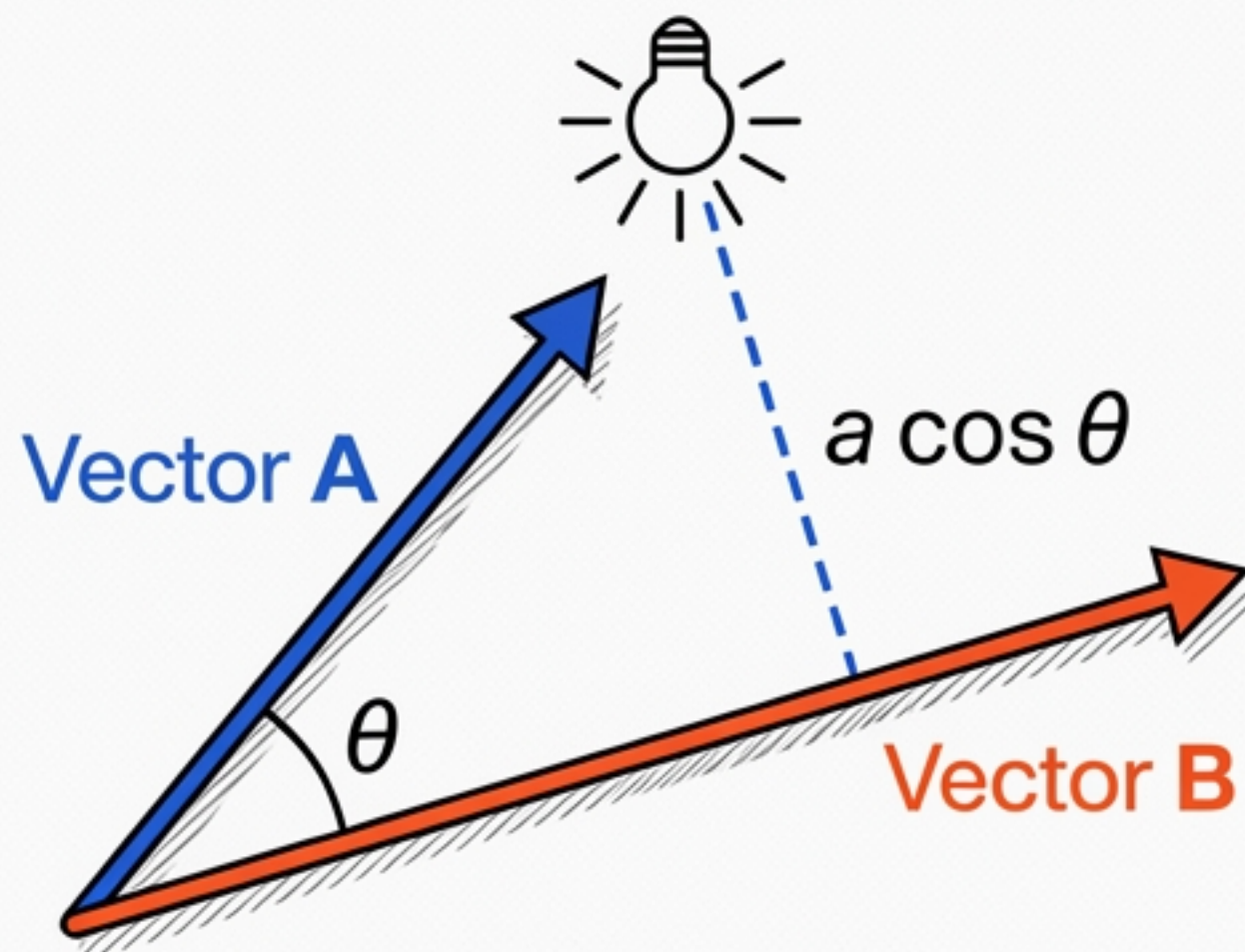
Example:

$$P(3, 4) \rightarrow d = \sqrt{3^2 + 4^2} = 5$$

Takeaway: In linear algebra, position and magnitude are interchangeable concepts depending on your reference frame (the origin).

Interaction: The Dot Product

Measuring Alignment



Definition: The **Dot Product** measures how much one vector goes in the **direction** of another.

Geometric Formula: $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta$

Note: The result is a SCALAR (a single number), not a vector.

Calculating the Dot Product

The mechanics

Algebraic Formula:

The sum of element-wise products.

$$\mathbf{A} \cdot \mathbf{B} = \sum a_i b_i = a_1 b_1 + a_2 b_2 + \dots$$

The rules

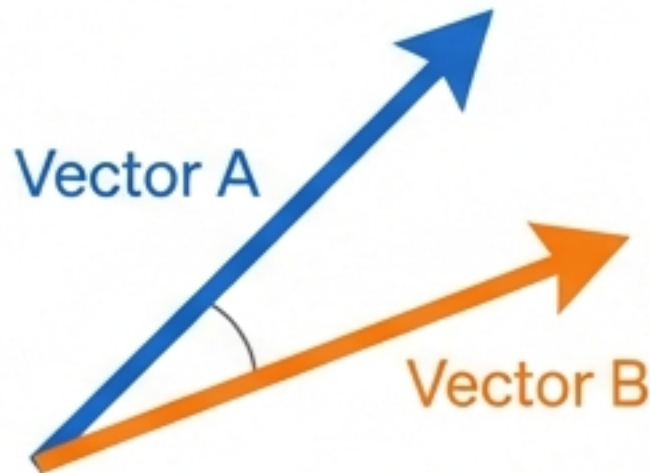
Key Properties:

1. Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
(Order does not matter)
2. Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
3. Parallel: If $\theta = 0^\circ$, result is $|\mathbf{A}||\mathbf{B}|$
(Maximum possible value)

Interpreting the Result

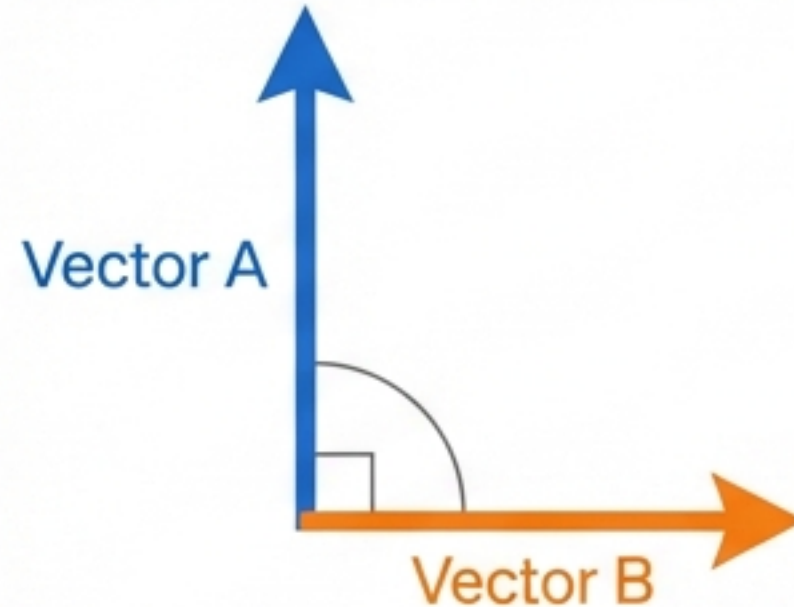
The sign tells you the relationship: aligned, orthogonal, or opposing.

Positive Result (+)



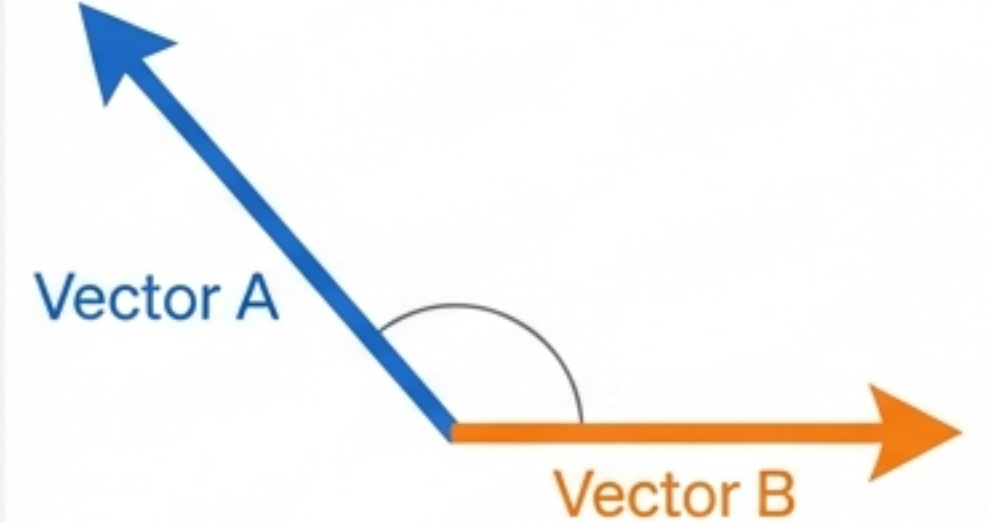
Generally Aligned.

Zero (0)



Perpendicular
(Orthogonal).
Crucial for checking
independence.

Negative Result (-)



Opposite Direction.

Dot Product in Practice

Manual Calculation

Scenario: $A = (3, 4, 5)$, $B = (2, -1, 3)$

Step 1: Multiply terms

$$(3)(2) + (4)(-1) + (5)(3)$$

Step 2: Sum results

$$6 - 4 + 15$$

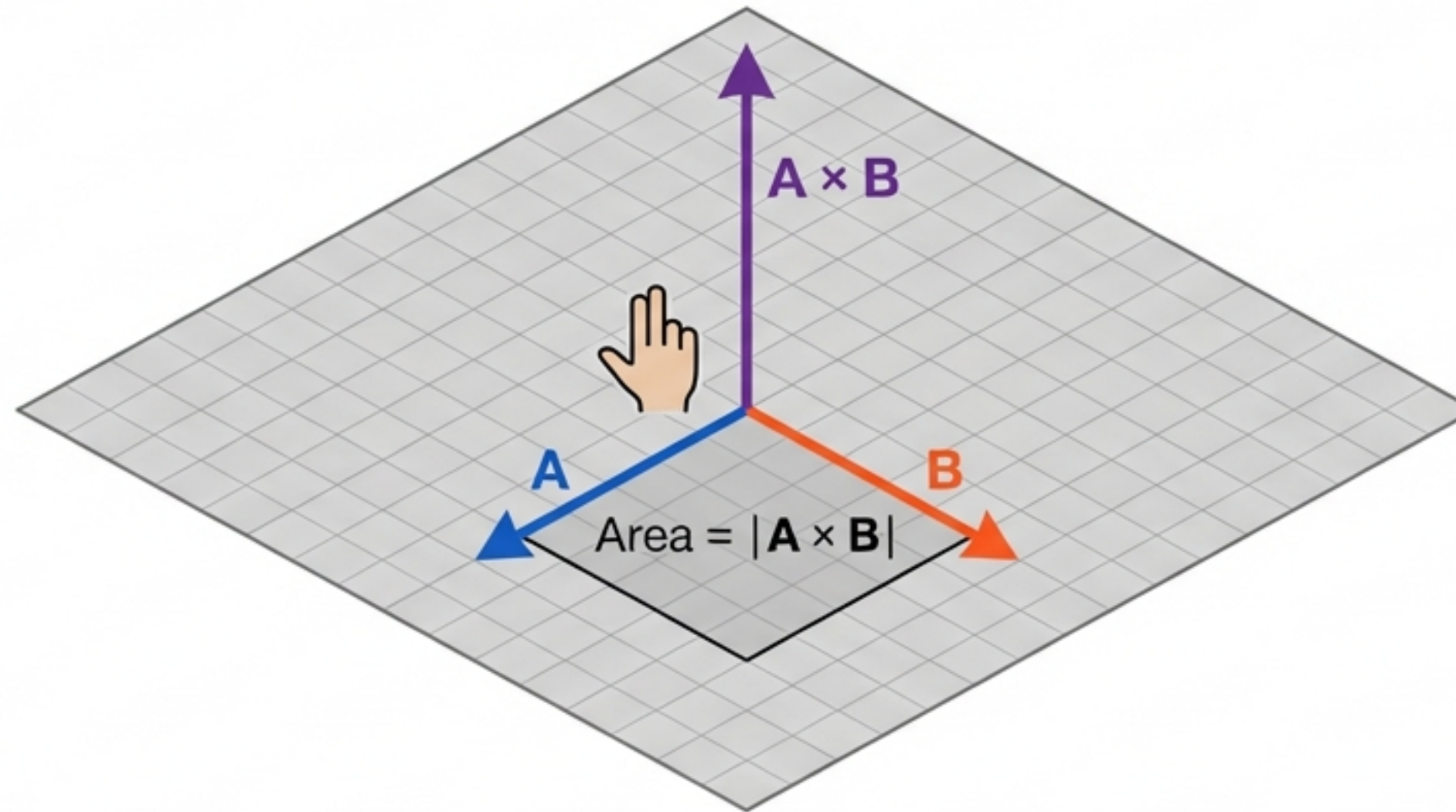
Result: 17

Python Implementation

```
import numpy as np
A = np.array([3, 4, 5])
B = np.array([2, -1, 3])
dot_product = np.dot(A, B)
print('Dot Product:', dot_product)
# Output: 17
```


Interaction: The Cross Product

Creating Orthogonality in 3D Space



Definition:

Creates a NEW VECTOR perpendicular to both input vectors.

Geometric Meaning:

The magnitude equals the area of the parallelogram formed by **A** and **B**.

Formula:

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \sin(\theta) \mathbf{n}$$

The Determinant Formula

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

i-component: $(A_y B_z - A_z B_y)$

$$\begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix}$$

j-component: $-(A_x B_z - A_z B_x)$

$$\begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix}$$

k-component: $(A_x B_y - A_y B_x)$

$$\begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

Properties of the Cross Product

1. **Non-Commutative: Order matters!**

$$\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A})$$

2. **Parallel Vectors:**

If $\mathbf{A} \parallel \mathbf{B}$, then $\mathbf{A} \times \mathbf{B} = \vec{0}$ (Zero Vector)

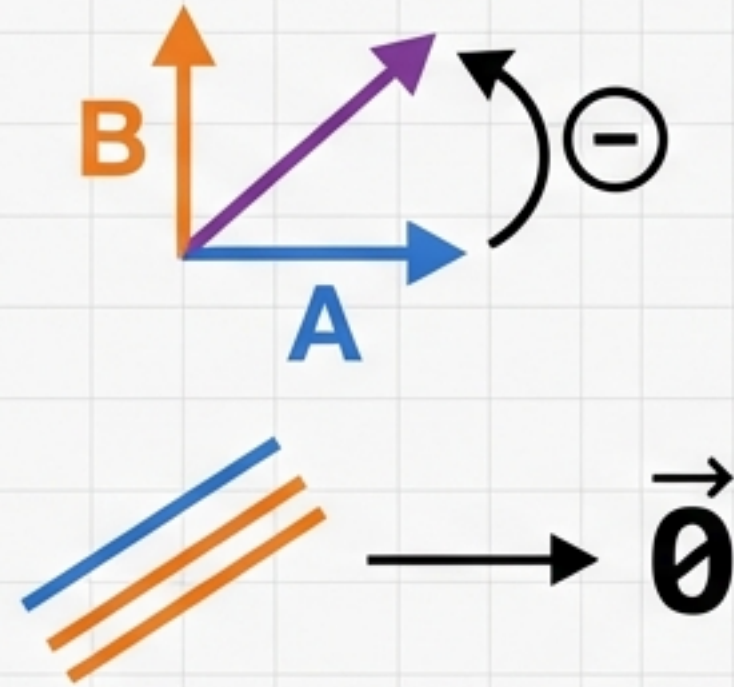
3. **Orthogonal Vectors:**

If $\mathbf{A} \perp \mathbf{B}$, magnitude is maximised ($|\mathbf{A}| |\mathbf{B}|$).

$$|\mathbf{A} \times \mathbf{B}| = |\mathbf{A}| |\mathbf{B}|$$

4. **Distributive:**

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$$



Cross Product: Step-by-Step Calculation

Input: $\mathbf{A} = (2, 3, 4)$, $\mathbf{B} = (5, 6, 7)$

1. Setup Matrix:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 4 \\ 5 & 6 & 7 \end{vmatrix}$$

2. Calculation:

$$\mathbf{i} (3 \cdot 7 - 4 \cdot 6) = \mathbf{i} (21 - 24) = -3\mathbf{i}$$

$$-\mathbf{j} (2 \cdot 7 - 4 \cdot 5) = -\mathbf{j} (14 - 20) = -(-6)\mathbf{j} = +6\mathbf{j}$$

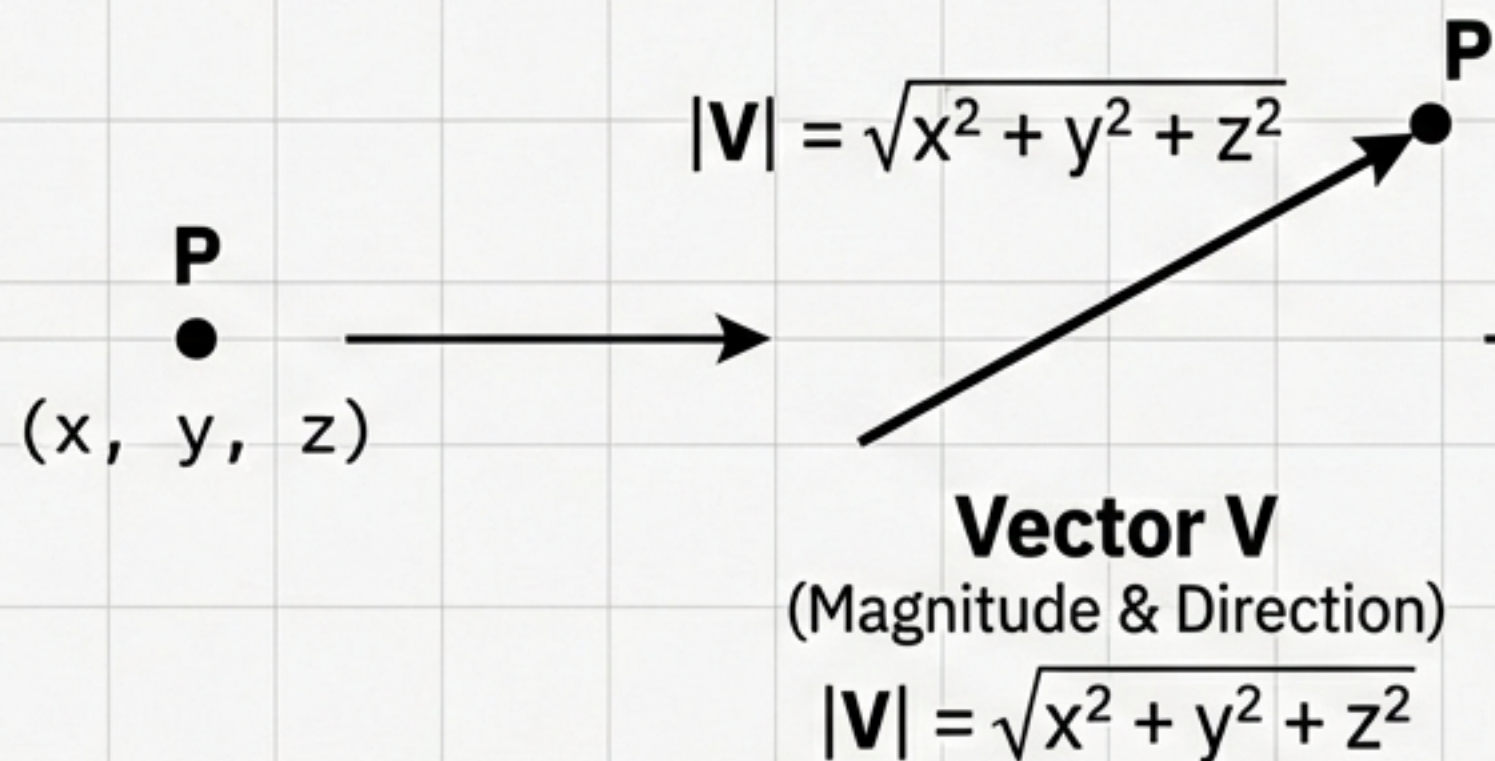
$$\mathbf{k} (2 \cdot 6 - 3 \cdot 5) = \mathbf{k} (12 - 15) = -3\mathbf{k}$$

3. Final Result Vector:
 $(-3, 6, -3)$

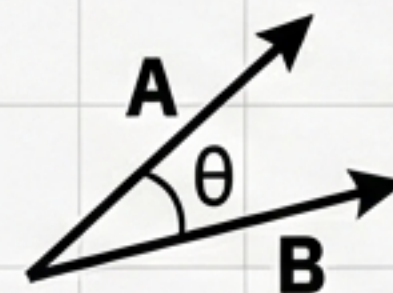
The Cheat Sheet: Dot vs. Cross

Feature	Dot Product ($\cdot$)	Cross Product ($\times$)
Result Type	Scalar (Number)	Vector
Geometric Formula	$ \mathbf{A} \mathbf{B} \cos \theta$	$ \mathbf{A} \mathbf{B} \sin \theta \cdot \mathbf{n}$
Measures	Alignment / Similarity	Area / Perpendicularity
Commutative?	Yes	No (Anti-commutative)
Zero Result	When Perpendicular	When Parallel

From Components to Complexity



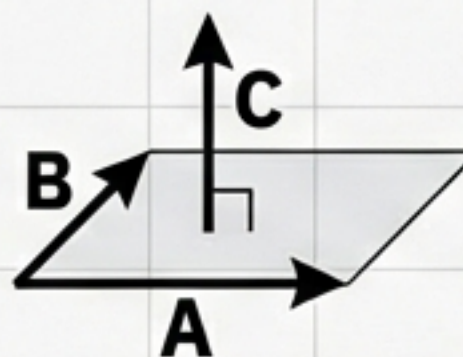
Interaction: Dot Product



$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta$$

Measures Alignment / Similarity

Interaction: Cross Product



$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}|\sin\theta \cdot \mathbf{n}$$

Constructs New Dimensions / Perpendicularity

We started with a single point in space. By defining magnitude and direction, we unlocked the ability to measure similarity and construct new dimensions.

These are the building blocks of spatial intelligence.